

Anurag Basavaraj Bhusare

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Technical Skills

AI/ML & Agents: OpenAI Agents SDK, LangChain, LangGraph, LangSmith, MCP, PyTorch, Transformers, RAG, PEFT, LoRA, Fine-tuning, Diffusion Models

Python Libraries: FastAPI, Gradio, Pydantic, Pandas, Ollama, Selenium

Cloud: Azure AI Stack (Video Indexer, AI Search, OpenAI Service, Monitor, OpenTelemetry), Docker

Databases: ChromaDB, FAISS, PostgreSQL, MySQL, Redis

Languages: Python, JavaScript, C++, SQL

Work Experience

BT Group

Jan 2025 – Present

Software Engineer

- Patched a critical Apache Tomcat vulnerability across 8 production servers within a 21-day SLA by reverse-engineering the CFML/Lucee/Tomcat stack, delivering a zero-downtime upgrade for 158K+ active users.
- Redesigned ServiceNow email workflows across 8 CIA products, eliminating plain-text credential exposure and improving security compliance without disrupting downstream notification dependencies.
- Built ITSM/ITAM workflows, Virtual Agent flows, and Flow Designer automations in ServiceNow, improving request handling efficiency for internal users and vendors.

Right Media Tech Private Limited

May 2024 – Jul 2024

GenAI Intern

- Built generative AI pipelines across diffusion models, voice AI, and RAG systems, reducing processing time by 40% and operational overhead by 25%.
- Developed workflow automation pipelines for AI content generation and retrieval, reducing manual intervention and improving operational efficiency.

Projects

Autonomous Trading Simulator

OpenAI Agents SDK, MCP, Gemini 2.5 Flash, Python asyncio

- A multi-agent trading simulator with four autonomous agents, enabling real-time market analysis and execution through Polygon.io APIs with persistent memory via LibSQL knowledge graph.
- Three-tier orchestration system with orchestrator, trader, and researcher agents, integrating 6 MCP servers for market data, account management, and web search.
- Implemented real-time observability dashboards using a custom TracingProcessor, exposing concurrent agent execution and tool traces in Gradio via asyncio.

Portfolio RAG Chatbot with Evaluation

Python, ChromaDB, Ollama, LangChain, Gradio

- Designed 4 modular RAG pipelines using Gemini, Qwen, ChromaDB, and LangChain, combining semantic chunking, query rewriting, vector search, and LLM reranking.
- Created an evaluation framework across 10 benchmark queries, measuring retrieval performance with MRR (0.585), nDCG (0.653), and 85.6% keyword coverage.
- Implemented a Pydantic-based LLM-as-a-judge to benchmark accuracy and relevance across Gemini-2.5-Flash (4.30/5, 4.80/5) vs. local Qwen3:4B (3.80/5, 4.30/5).

Video Compliance AI

Python, LangGraph, Azure Video Indexer, Azure AI Search, Azure OpenAI, FastAPI

- An agentic video compliance auditing system that analyzes ad videos against FTC regulations, generating structured compliance reports from YouTube URLs.
- Integrated Azure Video Indexer, Azure AI Search, and LangGraph to enable multimodal extraction and RAG-based compliance reasoning over regulatory documents.
- Deployed as a FastAPI service with observability via LangSmith, Azure Monitor, and OpenTelemetry for traceable compliance workflows.

LLM Price Prediction Engine

PyTorch, Llama 3.2, QLoRA, scikit-learn, HuggingFace

- Fine-tuned Llama 3.2 with QLoRA for product price prediction, achieving \$39.85 MAE — outperforming Gemini-2.5-Flash (\$62.84) and the human benchmark (\$87.62).
- Benchmarked 12 models across 5 approaches, including traditional ML, deep learning, zero-shot LLMs, and fine-tuned open-source LLMs.
- Engineered a scalable ML data pipeline for 400k+ Amazon products, combining parallel ingestion, LLM-based product summarization, and a reusable evaluation framework with error trend analysis at 95% CI.

Education

PES University, Bengaluru

Bachelor of Technology in Computer Science & Engineering | GPA: 8.8

Specialization: Data Science and Machine Learning

Jul 2025